



A discrete serotonergic circuit involved in the generation of tinnitus behavior

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Although dysregulated serotonergic neurotransmission has been implicated in the pathophysiology of tinnitus, the precise neural circuit mechanisms underlying this complex sensory neurological disorder remain elusive. In the current study, we investigated whether a serotonergic input from the dorsal raphe nucleus (DRN) to the dorsal cochlear nucleus (DCN), a key auditory brain region whose hyperactivity is associated with tinnitus, modulates behaviors in mice consistent with the presence of tinnitus. Using neural tracing and viral-genetic methods, we identified an anatomically and functionally defined serotonergic subpopulation in the DRN that projects to the DCN (5-HT^{DRN→DCN} neurons). Optogenetic activation of 5-HT^{DRN→DCN} circuit increased spike activity in DCN fusiform cells, exhibiting characteristics consistent with tinnitus-like electrical hyperactivity. Chemogenetic activation of 5-HT^{DRN→DCN} circuit induced tinnitus-related behaviors in mice, which was largely reversed by blocking 5-HT_{2A} receptors. Additionally, we found that noise exposure increased 5-HT levels in the DCN and the activity of 5-HT^{DRN→DCN} neurons in mice with noise-induced tinnitus-related behaviors. Importantly, chemogenetic inhibition of 5-HT^{DRN→DCN} circuit ameliorated significantly noise-induced tinnitus-related behavior in mice. These results reveal that activation of 5-HT^{DRN→DCN} circuit induces hyperactivity in the DCN sufficient for the perceptual generation and modulation of tinnitus. These findings provide direct evidence that 5-HT neurons in the DRN play an important role in tinnitus and facilitate our understanding of the circuit mechanisms of pathophysiology in sensory neurological disorders.

dorsal raphe nucleus | dorsal cochlear nucleus | 5-HT | tinnitus behavior | hyperactivity

Tinnitus, a perception of a phantom sound, is a common hearing disorder that affects approximately 10 to 15% of the global population (1). It is frequently accompanied by psychological symptoms, such as anxiety and depression (2, 3). Despite its high prevalence and significant impact on quality of life (4), the underlying mechanisms of tinnitus remain poorly understood.

Serotonin (5-hydroxytryptamine or 5-HT), a major neuromodulator widely distributed in the central nervous system (CNS), plays a key role in regulating a variety of essential neuronal functions, including emotional responses, sleep, cognitive processes, and sensory perception in the brain (5–8). Dysfunction of 5-HT signaling in the CNS is implicated in a wide range of psychiatric and neurological disorders, such as anxiety, depression, autism, pain, and migraine (9–11). While serotonergic pathways from the raphe nucleus to the limbic system have been extensively investigated for their roles in psychiatric diseases (6, 8, 10, 12–16), the functions of sensory-projecting 5-HT neural circuits and their potential contributions to the pathology of neurological disorders remain elusive.

Multiple lines of evidence implicate the 5-HT system in tinnitus (17–19). In clinical and preclinical studies, antidepressants such as selective serotonin reuptake inhibitors (SSRIs) have been explored as treatments for tinnitus in patients with and without depression (20–22). However, the success of such treatments is inconsistent, with some patients reporting the onset or exacerbation of tinnitus (23, 24). Moreover, evidence from animal studies suggests that dysregulated 5-HT signaling in the central auditory system is associated with tinnitus (19, 25, 26). For example, 5-HT neurons are activated during salicylate-induced tinnitus in gerbils (19). Loud noise exposure, which can cause tinnitus, may upregulate extracellular 5-HT levels and decrease serotonin transporter (SERT) expression in the central auditory system (25, 26). Together, these observations suggest that dysfunction of 5-HT signaling in the central auditory system is closely linked to the generation and perception of tinnitus. However, the underlying circuitry mechanisms by which 5-HT dysfunction contributes to the pathogenesis of tinnitus remain unclear.

Significance

Tinnitus, a prevalent hearing disorder linked to dysregulated serotonergic system, has poorly understood neural circuits. Using cutting-edge neuroscience tools, our study reveals a discrete 5-HT^{DRN→DCN} circuit driving tinnitus in hyperserotonergic states. These findings highlight serotonin's role in modulating sensory brain regions and identify circuit-level mechanisms of neurological disorders. Our work advances understanding of tinnitus and suggests targeted interventions.

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The dorsal cochlear nucleus (DCN) is a brain region involved in sound localization and has been strongly implicated in the triggering and modulation of tinnitus (4, 27–33). It receives dense serotonergic inputs primarily from the dorsal raphe nuclei (DRN) (26, 34–40). In addition to serotonergic innervation, several subtypes of 5-HT receptors, including 5-HT_{2A} receptor, are found in the cochlear nucleus (CN) (36). Our previous studies have shown that 5-HT directly excites fusiform cells, the principal output neurons of the DCN, leading to electrical hyperactivity (increased spontaneous firing rates, a tinnitus-like electrical behavior) in the DCN (40, 41). Furthermore, several studies suggest that noise exposure upregulates extracellular 5-HT levels in the DCN (25), and downregulates SERT expression (26), indicating a dysfunction of 5-HT signaling in the DCN. Given that both dysfunctions of the 5-HT system and the DCN are implicated in tinnitus generation or perception, we hypothesize that the dysfunction of DRN→DCN serotonergic (5-HT^{DRN→DCN}) circuit may play a role in the mechanisms underlying tinnitus.

We tested this hypothesis by using circuit tracing, viral-genetic tools, fiber photometry, electrophysiology, behavioral testing, and optogenetic/chemogenetic manipulations to systematically dissect the 5-HT^{DRN→DCN} circuit and elucidate its role in regulating neuronal activity of the DCN fusiform cells and driving behavior in mice indicative of tinnitus.

Results

Defining Connections of 5-HT^{DRN→DCN} Circuit in Mice. The CN is extensively innervated by serotonergic fibers in a variety of species, including rats and guinea pigs (37, 42), and this serotonergic innervation originates predominantly in the DRN (35, 37, 38). The DRN can be divided into several distinct subregions based on anatomy, connectivity, and functional topography, including the rostral, dorsal, ventral, ventrolateral, interfascicular, and caudal parts (6). However, the specific source of serotonergic projections to the DCN in mice has not yet been delineated at the subregional level within the DCN. By combining retrograde tracing with immunofluorescence staining for tryptophan hydroxylase 2 (TPH2), we aimed to identify the sources of serotonergic innervation in the DCN of mice. Green retrobeads were stereotactically injected into the DCN, either unilaterally or bilaterally (Fig. 1, *SI Appendix*, Fig. S1, and *Dataset S1 A and G*), resulting in retrograde labeling of neurons in several raphe nuclei (Fig. 1 *B and C*). The majority of the bead-containing neurons were also labeled for TPH2 and were found in the caudal DRN (DRc), interfascicular DRN (DRi), dorsal DRN (DRd), ventral DRN (DRv), and median raphe nucleus (MnR) (Fig. 1 *C*). The highest number of double-labeled neurons projecting to the DCN was found in the DRc. The double-labeled neurons in the DRc accounted for an average of 66.5% of the serotonergic neurons projecting to the DCN. The remaining serotonergic neurons were distributed as follows: 11.8% in the DRi, 14.6% in the DRd, 1.9% in the DRv, and 5.2% in the MnR (Fig. 1 *E*).

To further confirm the 5-HT^{DRN→DCN} circuit, we injected AAV-DIO-EGFP virus into the DRN of Pet1-Cre transgenic mice, which express Cre recombinase driven by the PET1 enhancer element in the raphe nuclei (43). Three weeks after the viral injection, we observed EGFP-labeled neurons in the DRN (Fig. 1 *G and H*). The majority of these EGFP-labeled neurons in the DRN were also labeled for TPH2 (Fig. 1 *H*), confirming successful viral infection leading to expression of EGFP in serotonergic neurons. Importantly, EGFP-labeled fibers were densely distributed across all three subregions of the DCN (Fig. 1 *G*), confirming that the

DCN receives direct projections from serotonergic neurons in the DRN.

In order to determine if serotonergic fibers terminate near glutamatergic DCN fusiform cells, we injected AAV-DIO-mCherry virus into the DCN of Vglut2-Cre mice and conducted immunostaining for SERT (Fig. 1 *I and J*). mCherry-labeled neurons were indeed surrounded by SERT-positive fibers and varicosities (Fig. 1 *K*). Our results revealed that serotonergic fibers likely form synaptic sites near glutamatergic neurons, which are most likely the fusiform principal cells in the DCN.

Taken together, our retrograde and anterograde double-labeling experiments demonstrated that a subpopulation of serotonergic neurons in the DRN is the primary source of serotonergic afferents to the DCN (Fig. 1 *L*).

Activation of 5-HT^{DRN→DCN} Circuit Evokes Hyperactivity of Fusiform Cells. Our previous study showed that exogenous 5-HT, or 5-HT release, can directly increase the excitability of fusiform cells, as demonstrated using brain-slice patch-clamp recordings (40, 41). To investigate whether activation of the 5-HT^{DRN→DCN} circuit affects the function of fusiform cells, both in vitro and in vivo analyses, we injected Cre-dependent AAV expressing the ChR2-tdTomato into the DRN of Pet1-Cre mice (Fig. 2 *A*). This optogenetic approach enabled the selective expression of ChR2-tdTomato in serotonergic neurons within the DRN, allowing for precise photostimulation of these neurons. Using immunohistochemical staining, we observed robust ChR2-tdTomato expression in the DRN (Fig. 2 *B*). Additionally, we confirmed that Cre activity in Pet1-Cre mice was selectively restricted to serotonergic neurons, as identified by TPH2 immunoreactivity (Fig. 2 *B*). To confirm the functional expression of ChR2 in 5-HT neurons in ChR2-tdTomato mice, we performed electrophysiological recordings of ChR2-tdTomato-expressing neurons in the DRN using acute brain slices. Serotonergic neurons were identified based on their characteristic large polymorphic somata, location, and tdTomato expression in soma (Fig. 2 *C and D*). Under voltage clamp recording, we observed photostimulation-induced currents in the tdTomato-expressing neurons (Fig. 2 *E* and *SI Appendix*, Fig. S2). Trains of blue light evoked reliably firing of action potentials in DRN 5-HT neurons at frequencies up to 20 Hz, as demonstrated under current-clamp recordings (Fig. 2 *E and F*, *SI Appendix*, Fig. S2, and *Dataset S1B*). These results confirm the functional ChR2 selectively expressed in 5-HT neurons. We observed hyperactivity, characterized by a tinnitus-like electrical activity in fusiform cells, induced by applied 5-HT (Fig. 2 *G*). This finding aligns with our previous studies (40, 41), further supporting the role of 5-HT in promoting neural hyperactivity associated with tinnitus-like phenomena.

To determine whether this 5-HT^{DRN→DCN} circuit modulates the function of fusiform cells in vivo, we implanted a fiber optic cannula above the DRN and conducted extracellular single-unit recordings from putative fusiform cells in awake, behaving Pet1-Cre::ChR2^{DRN→DCN} mice (Fig. 2 *H*). Histological analysis revealed the viral injection and expression sites, as well as the trajectory of the optical fiber tract positioned above the DRN (Fig. 2 *I*). Putative fusiform cells were identified based on their distinct physiological characteristics, including single-unit recordings that exhibited build-up poststimulus time histograms (PSTH), nonmonotonic rate-intensity functions at characteristic frequency (CF), and responses to broadband noise (44) (Fig. 2 *J–L*). We applied photostimulation (473 nm, 20 Hz) to the DRN and assessed the effects of optogenetic stimulation of 5-HT^{DRN→DCN} circuit on the firing activity of fusiform cells. Blue light

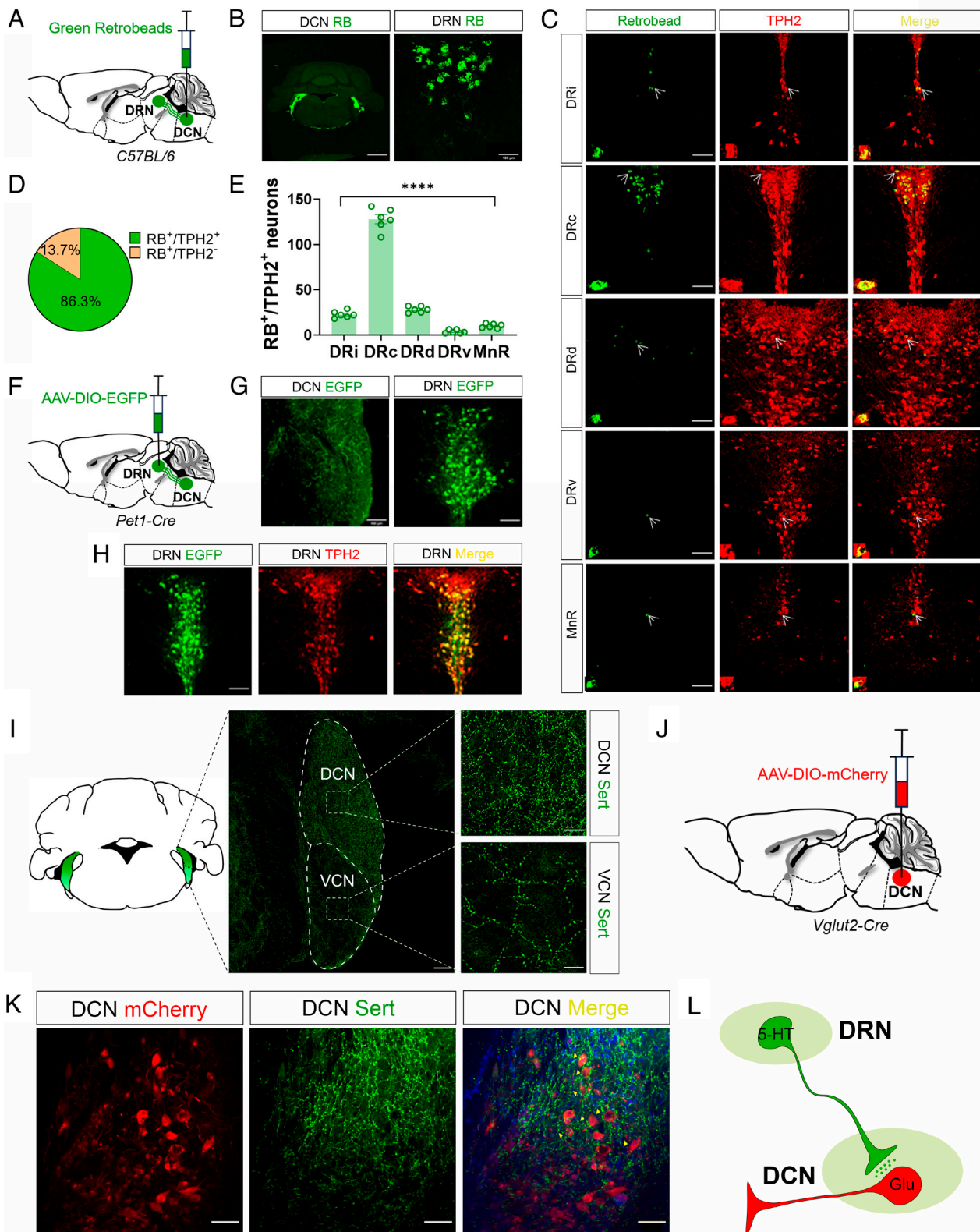


Fig. 1. Dissection of the 5-HT^{DRN-DCN} circuit. (A) Schematic illustrating the injection site of retrobeads in the DCN. (B) Representative images showing retrobeads expression in the DCN and DRN, respectively. (Scale bars: Left, 1,600 μ m; Right, 100 μ m.) (C) Representative images showing coexpression of retrobeads and TPH2 in different subregions of the DRN. (Scale bar: 200 μ m.) (D) Pie chart depicting the percentage of retrobeads-positive (RB⁺) cells coexpressing TPH2 (RB⁺/TPH2⁺) and those not coexpressing TPH2 (RB⁺/TPH2⁻). (E) Quantitative analysis of retrobeads expression in different subregions of the DRN ($n = 6$, **** $P < 0.0001$, one-way ANOVA). (F) Schematic illustrating the injection site of the virus in the DRN of Pet1-Cre mice. (G) Representative images showing virus expression in the DCN and DRN, respectively. (Scale bar: 100 μ m.) (H) Representative images showing coexpression of EGFP and TPH2 in the DRN. (Scale bar: 200 μ m.) (I) Representative images comparing SERT expression between the DCN and VCN. (Scale bars: Left, 200 μ m; Right, 20 μ m.) (J) Schematic illustrating the injection site of AAV-DIO-mCherry in the DRN of Vglut2-Cre mice. (K) Representative images showing coexpression of mCherry and SERT in the DRN. (Scale bar: 50 μ m.) (L) Schematic model of the 5-HT^{DRN-DCN} neural circuit.

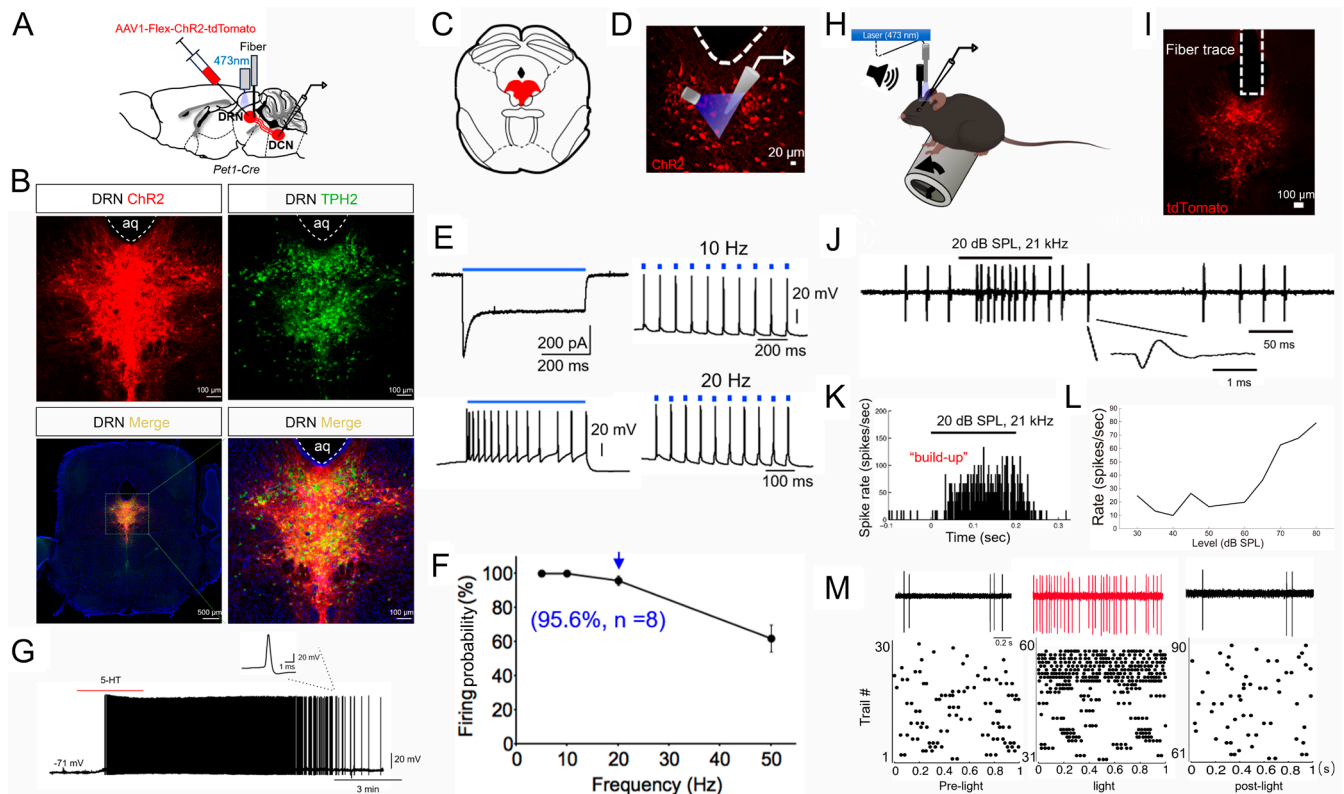


Fig. 2. Optogenetic activation of the 5-HT^{DRN-DCN} circuit triggers hyperactivity in putative fusiform cells. (A) Schematic representation of viral injection in Pet1-Cre mice. (B) Representative images showing coexpression of ChR2 and TPH2 in the DRN. (C and D) Schematic representation of the recording configuration in acute brain slices. (E, Left) Sustained light pulses (0.5 s) elicited a strong inward current in the voltage-clamp mode (Upper) and injection of +200 pA current evoked action potential firing in the current-clamp mode (Lower). (E, Right) Spike firing evoked by trains of brief light pulses (2 ms duration) in the current-clamp mode. (F) The mean ratio of successful spike generation in relationship to light stimulation at different frequencies ($n = 8$). (G) Applied 5-HT evoked a hyperactivity (increased spontaneous spike firing) of fusiform cells in the current-clamp mode. (H) Schematic diagram illustrating the positions of optic fiber implantation in the DRN and glass micropipettes in the DCN. (I) Outline of optical fiber and the expression of ChR2-tdTomato in the DRN. (J) Extracellular single-unit recording of a putative fusiform cell in response to 20 dB SPL 21 kHz pure tone. (K and L) Physiological responses of a fusiform cell to sound exhibit “build-up” poststimulus time histograms (PSTH, Left) and nonmonotonic rate-intensity functions at characteristic frequency (CF, Right). (M) Representative traces (Top) and the raster plots (Bottom) for spontaneous spike activity recorded in a putative fusiform cell before, during, and after light stimuli.

stimulation at 20 Hz clearly increased the rate of spontaneous spike firing (Fig. 2M).

Consistent with the observation in brain slice recordings (40), our *in vivo* physiological recordings suggest that the activation of 5-HT^{DRN-DCN} neurons induces hyperactivity, a tinnitus-like physiological response in fusiform cells of the DCN.

Activation of 5-HT^{DRN-DCN} Circuit Induces Tinnitus-Related Behaviors in Mice. Given that the electrical hyperactivity of fusiform cells in the DCN is strongly correlated with behavioral evidence of tinnitus (4, 27–33), the above observation that activation of 5-HT^{DRN-DCN} circuit directly induces hyperactivity in fusiform cells led to the hypothesis that activation of this circuit may contribute to the generation or modulation of behaviors consistent with the presence of tinnitus (“tinnitus-related behaviors”). To test this hypothesis, we expressed the hM3Dq chemogenetic activator (6, 45, 46) in 5-HT^{DRN-DCN} circuit by bilateral microinjections of a retrograde-tracer virus rAAV(RETRO)-DIO-FLP into the DCN of Pet1-Cre mice, followed by injections of either rAAV-fDIO-hM3Dq-EGFP or control AAV virus (rAAV-fDIO-EGFP) into the DRN (Fig. 3A). Four weeks after the viral injection, immunohistochemistry analysis confirmed the expression of hM3Dq-EGFP or control vectors in DRN serotonergic neurons (Fig. 3B and C). Consistent with our retrograde and anterograde double-labeling experiments, which identified a subpopulation of 5-HT neurons in the DRN as the primary source of serotonergic afferents to the DCN (Fig. 1),

we observed hM3Dq-EGFP-labeled 5-HT neurons (colocalized with TPH2) in the DRN. To confirm DREADD-based activation of serotonergic neuronal activity, we evaluated the expression of c-fos, an immediately-early gene indicative of *in vivo* neuronal activation, in the DRN using immunostaining following injection of the DREADD ligand clozapine N-oxide (CNO) (Fig. 3D). Coronal sections of brain slices from rAAV-fDIO-hM3Dq-EGFP- or rAAV-fDIO-EGFP-injected Pet1-Cre mice, which received a single intraperitoneal (i.p.) injection of CNO 30 min before they were killed, were analyzed. As shown in Fig. 3D, robust c-fos immunoreactivity was observed in the DRN of Pet1-Cre mice injected with rAAV-fDIO-hM3Dq-EGFP following CNO injection, validating the functionality of hM3Dq expressed in 5-HT neurons. In contrast, only minimal c-fos immunoreactivity was detected in the DRN of Pet1-Cre mice injected with rAAV-fDIO-EGFP 30 min after CNO injection (Fig. 3E).

Several behavioral paradigms were used to assess whether mice were perceiving a subjective sound while at the same time control for confounding factors affecting hearing responsiveness. Gap detection, assessed through the gap-prepulse inhibition of the acoustic startle reflex (GPIAS) is a widely used behavioral test for evaluating possible tinnitus in rodents (22, 47, 48) (Fig. 3F) as it directly assesses the perception of silence (28, 31, 32). In a second paradigm, prepulse inhibition (PPI), the acoustic startle reflex was inhibited by a weak, nonstartling prepulse stimulus presented shortly before the startling sound (Fig. 3F). Normal PPI, even in mice with a GPIAS deficit, indicates normal hearing, suggesting

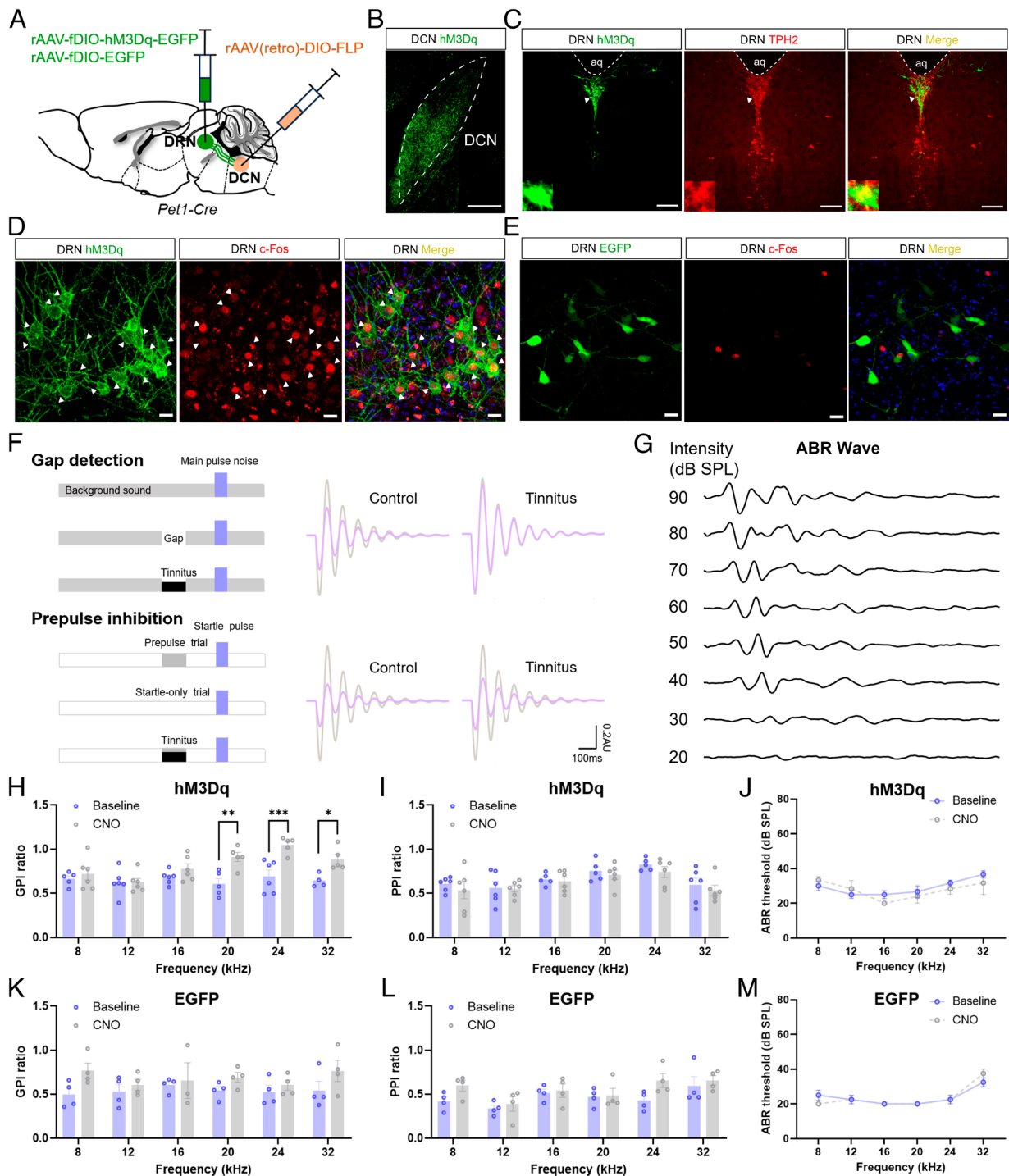


Fig. 3. Chemogenetic activation of the 5-HT^{DRN-DCN} circuit induces tinnitus-related behavior in mice. (A) Schematic diagram showing the injection sites of the chemogenetic virus rAAV-nEF1a-fDIO-hM3D(Gq)-EGFP-WPRE-PA or rAAV-nEF1a-fDIO-EGFP-WPRE-hGH polyA and a retrograde-tracing virus rAAV(RETRO)-DIO-FLP in the DRN and DCN of Pet1-Cre mice. (B and C) Representative images illustrating viral expression of hM3Dq-EGFP in the DCN and coexpression of hM3Dq-EGFP and TPH2 in the DRN. (Scale bar: 200 μ m.) (D and E) Representative images demonstrating coexpression of hM3Dq-EGFP and c-fos in the DRN and coexpression of EGFP and c-fos in the DRN. (Scale bar: 20 μ m.) (F, Top Left) Schematic of gap and no-gap trials in the gap detection behavioral assay. (Top Right) Representative startle responses in no-gap (gray) and gap trials (purple) for control and tinnitus mice. AU: Arbitrary units. (Bottom Left) Schematic of startle-only and prepulse trials in the PPI behavioral assay. (Bottom Right) Representative startle responses in startle-only (gray) and prepulse trials (purple) for control and tinnitus mice. (G) Representative raw traces of auditory brainstem response (ABR) to 24 kHz tone at varying intensities (dB SPL). (H) GPI ratio (response to gap trial/response to no-gap trial, with 8 to 32 kHz background noise) before (purple) and 7 d after CNO injection (gray) in hM3Dq-EGFP-expressing mice ($n = 4$ to 6; 8 kHz: $P = 0.9790$; 12 kHz: $P > 0.9999$; 16 kHz: $P = 0.7222$; 20 kHz: $P < 0.01$; 24 kHz: $P < 0.001$; 32 kHz: $P < 0.05$; two-way ANOVA). (I) Prepulse inhibition ratio (PPI ratio, response to prepulse trial/response to startle-only trial, with 8 to 32 kHz prepulse) before (purple) and 8 d after CNO injection (gray) in hM3Dq-EGFP-expressing mice ($n = 5$ to 6; 8 kHz: $P = 0.9454$; 12 kHz: $P > 0.9999$; 16 kHz: $P = 0.9999$; 20 kHz: $P = 0.9964$; 24 kHz: $P = 0.8997$; 32 kHz: $P = 0.9800$; two-way ANOVA). (J) Group hearing thresholds for each tested frequency before (purple line) and 8 d after (gray line) CNO injection in hM3Dq-EGFP-expressing mice ($n = 6$; NS, two-way ANOVA). (K) GPI ratio before (purple) and 7 d after CNO injection (gray) in EGFP-expressing mice ($n = 3$ to 4; 8 kHz: $P = 0.1774$; 12 kHz: $P = 0.9878$; 16 kHz: $P = 0.9994$; 20 kHz: $P = 0.7807$; 24 kHz: $P = 0.9904$; 32 kHz: $P = 0.3774$; two-way ANOVA). (L) PPI ratio before (purple) and 8 d after CNO injection (gray) in EGFP-expressing mice ($n = 4$; 8 kHz: $P = 0.3580$; 12 kHz: $P = 0.9945$; 16 kHz: $P > 0.9999$; 20 kHz: $P > 0.9999$; 24 kHz: $P = 0.1289$; 32 kHz: $P = 0.9857$; two-way ANOVA). (M) Group hearing thresholds for each tested frequency before (purple line) and 8 d after (gray line) CNO injection in EGFP-expressing mice ($n = 4$; NS, two-way ANOVA).

a GPIAS deficit is not due to hearing loss or general sensorimotor gating deficits. In addition, auditory brainstem response (ABR) was performed to rule out peripheral hearing loss. Gap detection performance can exhibit variability both among individual animals and across different groups. To account for this variability and ensure a more accurate assessment, we compared GPIAS before and after the injection of CNO within the same group (47, 49). One week of CNO treatment in hM3Dq-EGFP-expressing mice resulted in significantly impaired gap detection performance, as demonstrated by an increase in the GPI ratio (Fig. 3H and SI Appendix, Fig. S3). This deficit was reversible: following CNO withdrawal, the impairment showed significant recovery at 24 kHz within 1 d and was fully reversed across tested frequencies (16, 20, and 24 kHz) by day 3 (SI Appendix, Fig. S3D). By contrast, 1 wk of CNO treatment (1 mg/kg, i.p.) in hM3Dq-EGFP-expressing mice did not alter PPI or hearing thresholds as measured with ABR (Fig. 3I and J, SI Appendix, Fig. S3, and Dataset S1 C and H). Moreover, in control experiments, CNO injection alone did not alter GPIAS, PPI performance, or hearing thresholds (Fig. 3K–M), consistent with previous studies (47). Together, these results indicate that activation of 5-HT^{DRN→DCN} circuit is sufficient to induce tinnitus-related behaviors in mice.

5-HT_{2A} Receptors Are Implicated in Tinnitus-Related Behaviors in Mice Caused by Activation of 5-HT^{DRN→DCN} Circuit. Although multiple types of 5-HT receptors are expressed in the CN (36, 40)

and play a role in modulating the integration of auditory and nonauditory inputs, our previous studies indicated that the 5-HT_{2A} receptor primarily mediates the 5-HT-induced responses in fusiform cells of the DCN (41). To test whether 5-HT_{2A} receptors play a role in the tinnitus-related behavioral responses induced by activation of the 5-HT^{DRN→DCN} circuit, MDL-11939 (1 mg/kg, i.p.), a selective 5-HT_{2A} receptor antagonist, was administered to hM3Dq-EGFP-expressing mice before CNO injection (1 mg/kg, i.p.) and continued once daily for 7 consecutive days (Fig. 4 and Dataset S1D). Our results revealed that CNO treatment, followed by treatment with MDL-11939 in hM3Dq-EGFP-expressing mice, did not alter the GPI ratio (SI Appendix, Fig. S4 and Dataset S1I). This result suggests that inhibition of 5-HT_{2A} receptors attenuates the development of tinnitus caused by activation of the 5-HT^{DRN→DCN} circuit. Indeed, compared with hM3Dq-EGFP-expressing mice treated with CNO alone, treatment with MDL-11939 after CNO, but not vehicle, significantly reduced the percentage of hM3Dq-EGFP-expressing mice that developed GPI ratios suggestive of tinnitus down to a level comparable to that of control mice (Fig. 4G, control: 25.0%, CNO treatment: 83.3%, MDL-11939 + CNO treatment: 16.7%). Neither the drugs nor the viruses had any detectable effect on baseline auditory function, as measured by ABR thresholds, or on sensorimotor gating, as assessed by PPI performance (Fig. 4B, C, E, and F). These results indicate that activation of 5-HT^{DRN→DCN} circuits induces tinnitus-related behavior primarily mediated through 5-HT_{2A} receptors, supporting the hypothesis that the

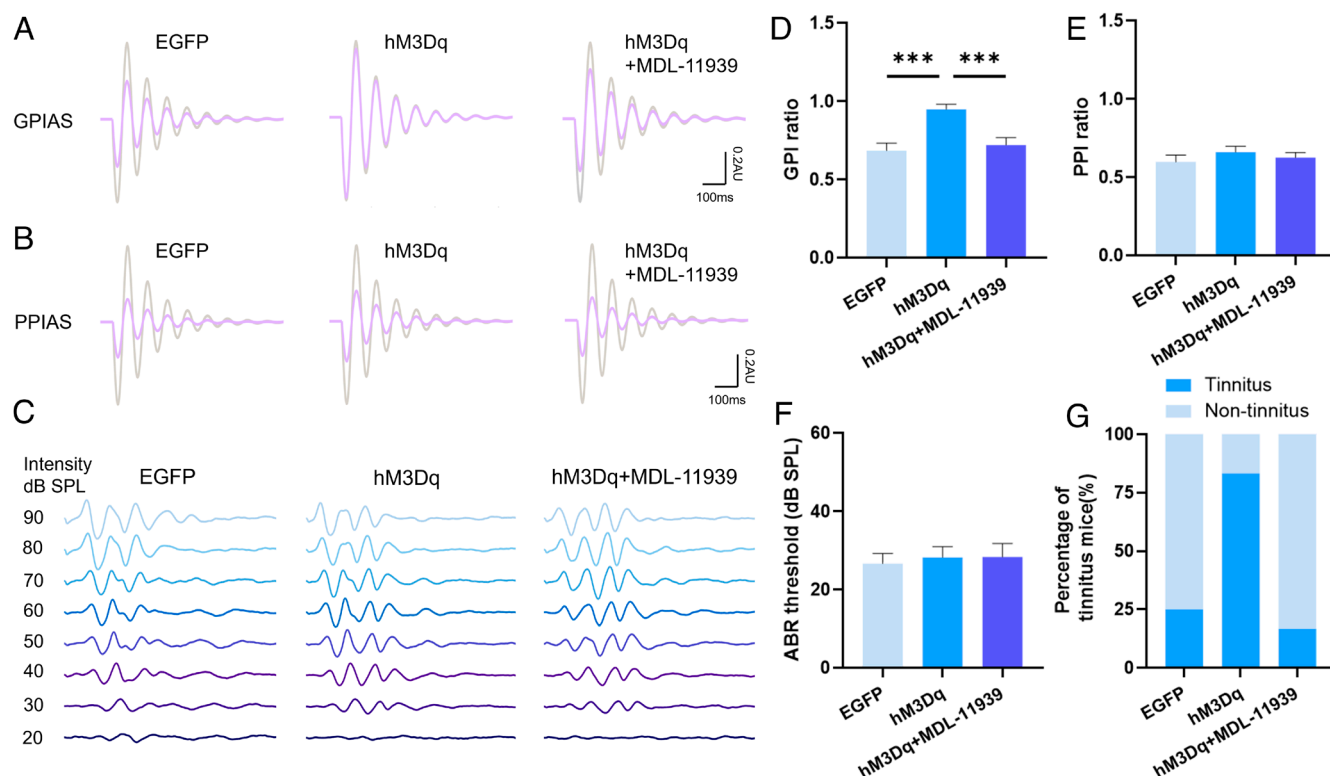


Fig. 4. The 5-HT_{2A} receptor antagonist reduces the incidence of tinnitus-related behavior induced by chemogenetic activation of the 5-HT^{DRN→DCN} circuit. (A) Representative startle responses in no-gap trials (gray) and gap trials (purple) for EGFP-, hM3Dq-, and hM3Dq+MDL-11939-injected mice. AU: Arbitrary units. (B) Representative startle responses in startle-only trials (gray) and prepulse trials (purple) for EGFP-, hM3Dq-, and hM3Dq+MDL-11939-injected mice. (C) Representative raw traces of ABR to 20 kHz stimuli at varying intensities (dB SPL) for EGFP-, hM3Dq-, and hM3Dq+MDL-11939-injected mice. (D) Summary graphs of GPI ratio for high-frequency background sounds (20 to 32 kHz; EGFP: *n* = 4, hM3Dq: *n* = 6, hM3Dq+MDL-11939: *n* = 6) 7 d after CNO or CNO + MDL-11939 injection (EGFP vs. hM3Dq: *P* < 0.001; hM3Dq vs. hM3Dq+MDL-11939: *P* < 0.001; two-way ANOVA). (E) Summary graphs of prepulse startle ratio (response to prepulse and startle stimulus/response to startle stimulus alone) for high-frequency prepulse sounds (20 to 32 kHz; EGFP: *n* = 4, hM3Dq: *n* = 6, hM3Dq+MDL-11939: *n* = 6). (F) ABR thresholds at test frequencies (8 to 32 kHz; EGFP: *n* = 4, hM3Dq: *n* = 6, hM3Dq+MDL-11939: *n* = 6). (G) Percentage of mice developing tinnitus (EGFP: 25%; hM3Dq: 83.3%; hM3Dq+MDL-11939: 16.67%).

5-HT^{DRN→DCN} circuit plays a pivotal role in the underlying mechanisms of tinnitus.

The 5-HT^{DRN→DCN} Circuit Is Activated in a Mouse Model of Noise-Induced Tinnitus. It is well established that exposure to loud noise can lead to the development of tinnitus in humans and tinnitus-related behaviors in animal models (28, 31, 32). Previous research has identified the DCN as a key brain region implicated in noise-induced tinnitus (27–29, 31, 32). Additionally, dysfunction of serotonergic system has been strongly associated with the generation or perception of tinnitus following acoustic trauma (17, 37). Notably, noise exposure, besides being a known trigger for tinnitus, also increases extracellular 5-HT levels in DCN (25). Therefore, we proposed that abnormal activity in 5-HT^{DRN→DCN} circuit may contribute to the development of noise-induced tinnitus. To test this possibility, we expressed a 5-HT sensor (5-HT3.0) in the DCN and performed fiber photometry-mediated fluorescence recording to directly measure the dynamics of serotonergic activity within the DCN of freely moving mice. This approach allowed us to compare serotonergic activity under quiet vs. noisy conditions (116 dB SPL white noise) (50), in mice with and without GPI ratios indicating tinnitus (Fig. 5 A–F and Dataset S1E). The noise exposure paradigm employed in this study has been previously demonstrated to induce tinnitus-related behavior in mice (28, 31, 32). Mice were exposed monaurally to either noise (116 dB SPL, 16 kHz, 2 h) or sham conditions, and tinnitus behavior was assessed using the GPIAS paradigm. We unilaterally injected rAAV-hSyn-5-HT3.0 or control AAV virus (rAAV-hSyn-EGFP) into the DCN region of C57BL/6 mice to express 5-HT3.0 for monitoring 5-HT signals. One week later, optic fibers were implanted above the viral injection site and a fiber photometry system was used to record the fluorescence responses of the 5-HT sensor when 5-HT is locally elevated (Fig. 5A). Four weeks after the viral injection, immunofluorescence analysis confirmed the expression of 5-HT3.0 or control vectors in the DCN (Fig. 5B). Coronal brain sections revealed the viral injection and expression sites, along with an optical fiber tract positioned above the DCN. We observed a dramatic increase in the fluorescence signal within the DCN after loud noise exposure (Fig. 5 C–E), indicating that 5-HT levels remained dramatically elevated. Importantly, the increase in 5-HT levels within the DCN of noise-exposed mice was accompanied by the emergence of tinnitus-related behavior (Fig. 5 C–F, SI Appendix, Fig. S5A, and Dataset S1J). In contrast, no detectable change in fluorescence signal was observed in mice expressing EGFP in the DCN during noise exposure (Fig. 5C, SI Appendix, Fig. S5 B and C, and Dataset S1J).

To further assess the neural activity of 5-HT^{DRN→DCN} neurons in mice exhibiting tinnitus-related behaviors, we recorded real-time calcium dynamics via fiber photometry recording in DRN of freely moving mice expressing genetically encoded fluorescent Ca²⁺ indicator GCaMP6f in 5-HT^{DRN→DCN} neurons (51). We performed bilateral microinjections of a retrograde-tracing virus rAAV(retro)-DIO-FLP into the DCN of Pet1-Cre mice, followed by injections of either rAAV-fDIO-GCaMP6f or control AAV virus (rAAV-fDIO-EGFP) into the DRN, to selectively express the calcium sensor GCaMP6f in 5-HT^{DRN→DCN} neurons (Fig. 5G). An optic fiber was implanted to target the DRN 1 wk after the viral injection, which allowed in vivo monitoring of 5-HT^{DRN→DCN} population neural activity in noise induced-tinnitus mice (6, 52). Post hoc histological analysis confirmed the selective expression of GCaMP6f in 5-HT^{DRN→DCN} neurons (Fig. 5 H and I). Coronal brain sections at the Bregma levels revealed viral injection and expression sites (Fig. 5J). We observed a significant increase in the

fluorescence signal within the DRN after noise exposure (Fig. 5 J–L), indicating that the activity of 5-HT^{DRN→DCN} neurons remained elevated. Moreover, the increased activity in 5-HT^{DRN→DCN} neurons of noise-exposed mice was accompanied by the emergence of tinnitus-related behavior (Fig. 5 J–M and SI Appendix, Fig. S5D). In contrast, no significant change in fluorescence signals was observed in 5-HT^{DRN→DCN} neurons of EGFP-expressing control mice with tinnitus (Fig. 5J and SI Appendix, Fig. S5 E and F), confirming that the GCaMP6f signals reliably and accurately reflected neuronal activity. The results demonstrated that the increased activity in 5-HT^{DRN→DCN} neuron is closely associated with the presence of tinnitus-related behavior.

Additionally, we assessed changes in the expression of c-fos as a marker of in vivo neuronal activity. Using immunohistochemical microscopy, we analyzed noise exposure-induced changes in the overall signal intensity of c-fos staining within the DRN of mice. We found that, at 2 h, 3 d, and 7 d after noise exposure, the overall intensity of c-fos staining was increased compared to serotonergic neurons under sham conditions (SI Appendix, Fig. S6 and Dataset S1K). It is noteworthy that the c-fos expression was predominantly increased in TPH2-expressing neurons but was also elevated in some TPH2-negative neurons within DRN, suggesting that nonserotonergic neurons may also be activated by noise exposure.

In summary, our findings demonstrate that both 5-HT release and 5-HT^{DRN→DCN} neuronal activity in tinnitus mice are significantly elevated compared to control mice, suggesting that hyperactivity of the 5-HT^{DRN→DCN} circuit is strongly associated with noise-induced tinnitus in mice.

Inhibition of 5-HT^{DRN→DCN} Circuit Ameliorates Noise-Induced Tinnitus-Related Behavior in Mice. Given the above observations implicating dysfunction of 5-HT^{DRN→DCN} circuit in the perception of noise-induced tinnitus, we aimed to determine whether silencing 5-HT^{DRN→DCN} circuit modulates the perception of noise-induced tinnitus in mice. We therefore chemogenetically inhibited the 5-HT^{DRN→DCN} circuit and evaluated its effect on noise-induced tinnitus-related behavior in mice (Fig. 6 and Dataset S1F). To selectively inhibit 5-HT^{DRN→DCN} circuit in vivo, we performed bilateral microinjections of a retrograde-tracing virus rAAV(retro)-DIO-FLP into the DCN of Pet1-Cre mice, followed by injections of either rAAV-fDIO-hM4D(Gi)-EGFP or control AAV virus (rAAV-DIO-EGFP) into the DRN (Fig. 6A). The hM4Di, a Gi/o-coupled DREADD, has proven effective for suppressing neuronal activity and chemogenetically silencing serotonergic circuits in awake, behaving animals (53, 54). Four weeks after the viral injection, immunohistochemistry analysis confirmed the expression of hM4D(Gi)-EGFP or control vectors (EGFP) in DRN serotonergic neurons (Fig. 6 B and C). Pet1-Cre animals expressing hM4D(Gi)-EGFP or EGFP (control AAV virus) with tinnitus-related behavior were administered CNO through daily injections (1 mg/kg, i.p.), followed by assessment of tinnitus responses. As expected, we found that inhibition of 5-HT^{DRN→DCN} circuit in hM4D(Gi)-EGFP expressing mice via CNO injection significantly reduced the GPI ratio, without affecting PPI performance or ABR threshold (Fig. 6D, SI Appendix, Fig. S7A, and Dataset S1L). In contrast, CNO administration in EGFP-expressing mice failed to reverse noise-induced GPI ratio changes or alter PPI and ABR threshold (Fig. 6E, SI Appendix, Fig. S7B, and Dataset S1L). These results suggest that inhibition of 5-HT^{DRN→DCN} circuit partially reverses noise-induced impairments in gap detection, without affecting baseline gap detection or prepulse inhibition behavior. Overall, compared with EGFP-expressing mice, treatment with CNO in hM4Di-EGFP-expressing mice reduced significantly

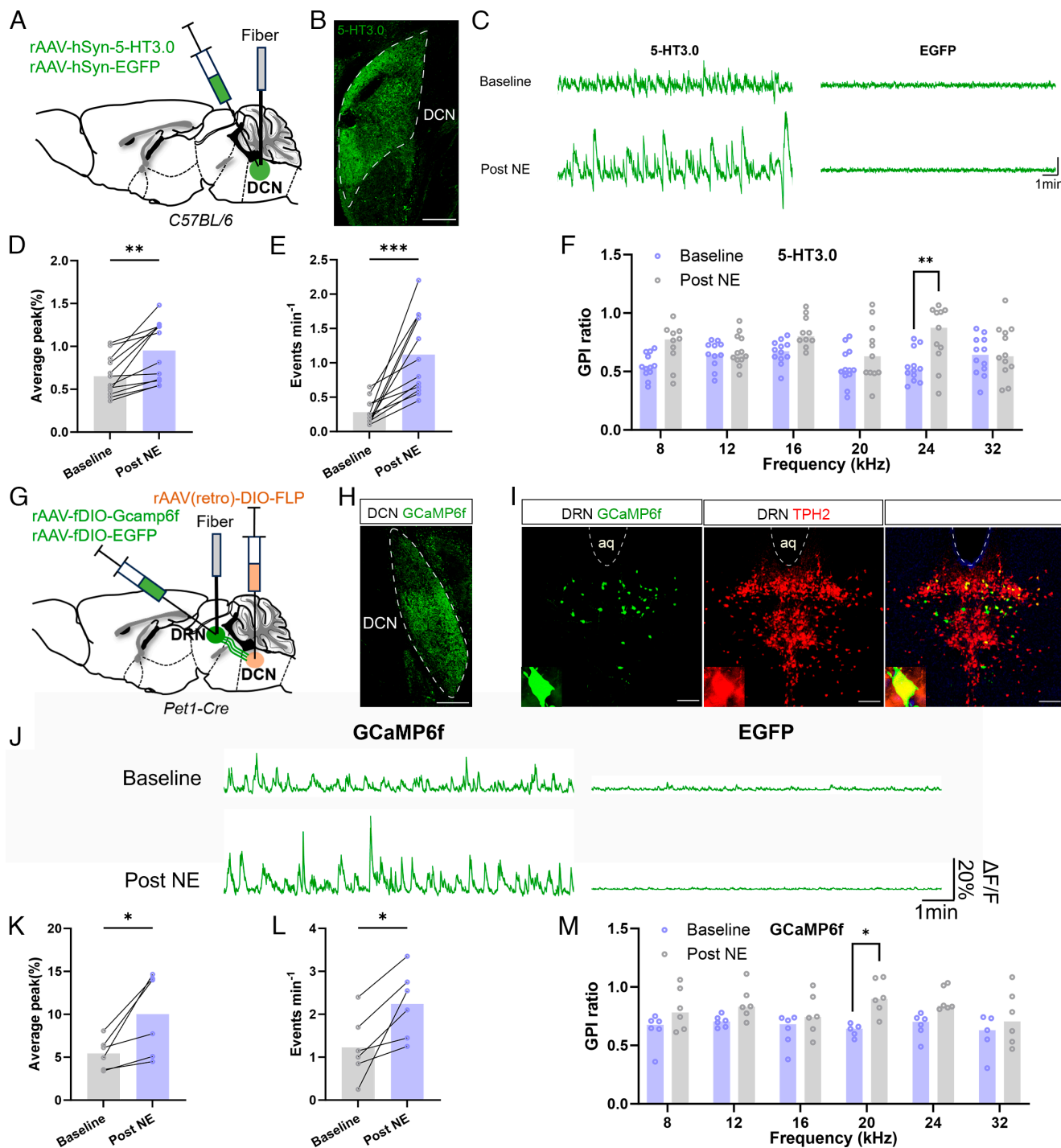


Fig. 5. Fiber photometry recording of 5-HT release in the DCN and 5-HT^{DRN→DCN} neural activity of mice before and after noise exposure (post-NE). (A) Schematic diagram of virus injection and recording sites (DCN). (B) Histological verification of 5-HT3.0 expression in the DCN. (C) Representative traces of 5-HT3.0 and EGFP fluorescence changes before and post-NE. (D) Average peak $\Delta F/F$ of 5-HT signals before and 6 d post-NE ($n = 12$; $P < 0.01$; paired t test). (E) Frequency of 5-HT signal events before and 6 d post-NE ($n = 12$; $P < 0.001$; paired t test). (F) GPI ratio in 5-HT3.0-expressing mice before and 7 d post-NE ($n = 10$ to 12, 24 kHz; $P < 0.01$, two-way ANOVA). (G) Schematic diagram of virus injection sites (DCN and DRN) and recording site (DRN). (H and I) Representative images showing viral expression of GCaMP6f in the DCN and coexpression of GCaMP6f and TPH2 in the DRN. (Scale bar: 200 μm .) (J) Representative traces of GCaMP6f and EGFP fluorescence changes before and post-NE. (K) Average peak $\Delta F/F$ of calcium signals before and 6 d post-NE ($n = 6$; $P < 0.05$; paired t test). (L) Frequency of calcium signal events before and 6 d post-NE ($n = 6$; $P < 0.05$; paired t test). (M) GPI ratio in GCaMP6f-expressing mice before and 7 d post-NE ($n = 5$ to 6, 20 kHz; $P < 0.05$, two-way ANOVA).

the percentage of mice showing tinnitus behavior (SI Appendix, Fig. S7C, hM4Di-EGFP with CNO treatment, 42.86% of control, $n = 7$; EGFP with CNO treatment, 100.0% of control, $n = 5$).

Taken together, these results reveal that aberrant activity of 5-HT^{DRN→DCN} circuit contributes to the pathology of noise-induced tinnitus (SI Appendix, Fig. S7D).

Discussion

Dysregulation of serotonergic neurotransmission has long been implicated in the pathophysiology of tinnitus. However, direct evidence linking the serotonergic circuit to the generation and modulation of tinnitus remains lacking. In this study, we identified a subpopulation of DRN neurons projecting to the DCN using

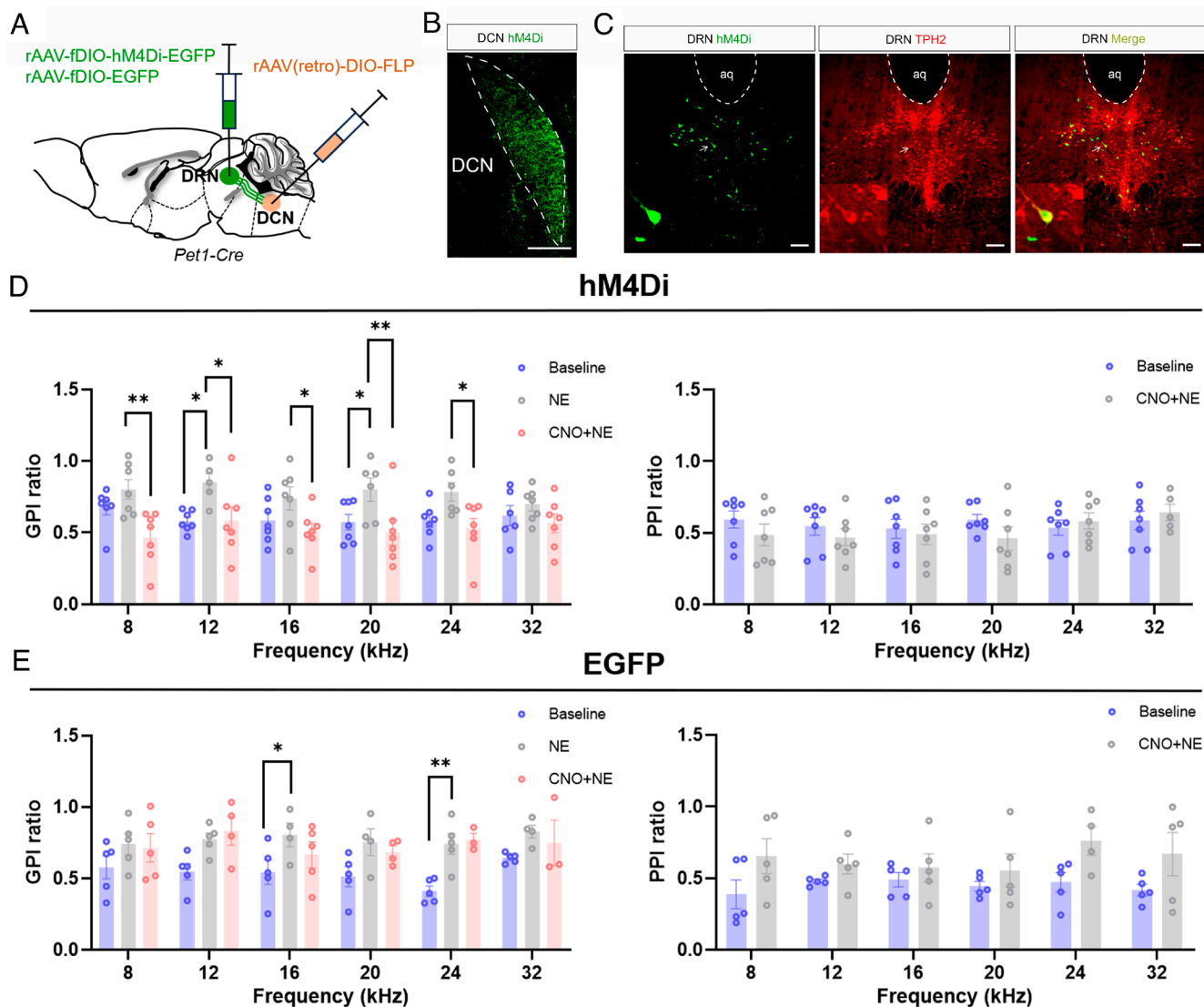


Fig. 6. Chemogenetic inhibition of the 5-HT^{DRN→DCN} circuit alleviates tinnitus-related behavior induced by noise exposure (NE). (A) Schematic diagram showing the injection sites of the chemogenetic virus rAAV-nEF1α-fDIO-hM4Di(Gi)-EGFP-WPRE-hGH pA or rAAV-nEF1α-fDIO-EGFP-WPRE-hGH polyA and a retrograde-tracing virus rAAV(RETRO)-DIO-FLP in the DRN and DCN. (B and C) Representative images showing viral expression in the DCN and coexpression of hM4Di-EGFP and TPH2 in the DRN. (Scale bar: 200 μm.) (D, Left) GPI ratio before (purple), 7 d after NE (gray), and after CNO injection (pink) in hM4Di-EGFP-expressing mice. Comparisons between NE and baseline: $n = 5$ to 7 ; 8 kHz: $P = 0.3063$; 12 kHz: $P < 0.05$; 16 kHz: $P = 0.1823$; 20 kHz: $P < 0.05$; 24 kHz: $P = 0.0649$; 32 kHz: $P = 0.6664$; two-way ANOVA. Comparisons between NE and CNO injection: $n = 5$ to 7 ; 8 kHz: $P < 0.01$; 12 kHz: $P < 0.05$; 16 kHz: $P < 0.05$; 20 kHz: $P < 0.01$; 24 kHz: $P < 0.05$; 32 kHz: $P = 0.2853$; two-way ANOVA. (D, Right) PPI ratio before (purple) and after CNO injection (gray) in hM4Di-EGFP mice ($n = 5$ to 7 ; 8 kHz: $P = 0.7920$; 12 kHz: $P = 0.9510$; 16 kHz: $P = 0.9986$; 20 kHz: $P = 0.6016$; 24 kHz: $P = 0.9960$; 32 kHz: $P = 0.9929$; two-way ANOVA). (E, Left) GPI ratio before (purple), 7 d after noise exposure (pink), and after CNO injection (gray) in EGFP-expressing mice. Comparisons between NE and baseline: $n = 4$ to 5 ; 8 kHz: $P = 0.2495$; 12 kHz: $P = 0.0726$; 16 kHz: $P < 0.05$; 20 kHz: $P = 0.0728$; 24 kHz: $P < 0.01$; 32 kHz: $P = 0.2057$; two-way ANOVA. Comparisons between NE and CNO injection: no significant differences, two-way ANOVA. (E, Right) PPI ratio before (purple) and after CNO injection (gray) in EGFP-expressing mice ($n = 4$ to 5 ; 8 kHz: $P = 0.7920$; 12 kHz: $P = 0.9510$; 16 kHz: $P = 0.9986$; 20 kHz: $P = 0.6016$; 24 kHz: $P = 0.9960$; 32 kHz: $P = 0.9929$; two-way ANOVA).

circuit tracing and viral-genetic approaches. We found that activation of the 5-HT^{DRN→DCN} circuit produced a hyperactivity of fusiform cells, a tinnitus-like physiological activity, as demonstrated through optogenetics and in vitro & in vivo physiological recordings (40). Remarkably, chemogenetic activation of 5-HT^{DRN→DCN} circuit was sufficient to reversibly induce behavior in mice that were consistent with a tinnitus percept. Furthermore, using fiber photometry recordings, we observed increased 5-HT release within the DCN and increased activity of 5-HT^{DRN→DCN} neurons in a noise-induced tinnitus mice model. Importantly, chemogenetic silencing of 5-HT^{DRN→DCN} circuit alleviated noise-induced tinnitus-related behavior in mice. Taken together, these findings provide compelling evidence that the 5-HT^{DRN→DCN} circuit regulates the excitability of the DCN and plays a critical role in the generation and modulation of tinnitus.

Given that the DRN consists of several distinct subpopulations of 5-HT neurons (6), identifying the specific source of 5-HT projections to the DCN is necessary for understanding serotonergic modulation of auditory function. Using neural tracing combined with viral-genetic methods, we identified a subpopulation of serotonergic neurons within the DRN that exhibit robust axonal projections to the DCN in mice (Fig. 1), confirming and extending previous studies in other species, such as cat and rat, which established the DRN as the major source of serotonergic inputs to the DCN (26, 34–40, 42). Consistent with previous anatomical studies, serotonergic fibers exhibit denser innervation of the DCN compared to the VCN or other auditory nuclei across multiple species (Fig. 1), reflecting a conserved organizational pattern (26, 35, 37, 38). The strong and preferential serotonergic innervation of the DCN suggests that 5-HT^{DRN→DCN} circuit plays

a fundamental role in modulating DCN circuitry. Our immunohistochemical analysis further revealed that serotonergic fibers localize preferentially near fusiform cells within the DCN, suggesting direct synaptic contacts (Fig. 1 J and K). However, the extensive serotonergic innervation throughout all DCN layers revealed by our viral tracing and SERT staining contrasts with the more restricted pattern observed in our previous study (40). This discrepancy likely arises from methodological sensitivity that viral tracing robustly labels complete axonal arbors and detects sparse terminals that were not captured by anti-GFP immunohistochemistry in Tph2-Chr2-YFP mice. Overall, our findings align with previous studies in guinea pigs and rats, which similarly identified serotonergic fiber apposition to fusiform cells, implying that serotonergic afferents may directly modulate fusiform cell activity in the DCN (36, 37, 42). Indeed, our in vitro and in vivo physiological experiments demonstrated that optogenetic activation of serotonergic terminals or 5-HT^{DRN→DCN} circuit robustly increases the spontaneous spike rate of fusiform cells, effectively a tinnitus-like hyperactivity in neurons conveying sound information to higher brain regions. This effect was reversed upon cessation of light stimulation and was mediated largely by the 5-HT_{2A} receptor (Fig. 2 and SI Appendix, Fig. S2). Similarly, in vitro brain slice electrophysiology demonstrated that both exogenous 5-HT and evoked 5-HT release robustly enhance DCN circuit excitability, with a pronounced hyperactivity of fusiform cells (41). It is well established that the hyperactivity in fusiform cells of the DCN is strongly associated with tinnitus (4, 27–33). Therefore, the hyperactivity of fusiform cells induced by 5-HT^{DRN→DCN} circuit most likely represents a tinnitus-like phenomenon.

Our results revealed that the chemogenetic activation of 5-HT^{DRN→DCN} circuit induces a mid- to high-frequency tinnitus phenotype, as evidenced by gap detection deficit (Fig. 3). We propose two mechanisms for the frequency-specific effect. First, 5-HT engages tonotopically organized inhibitory interneurons (55), and our data show this enhances inhibition of fusiform cells, disproportionately suppressing low-frequency regions (SI Appendix, Fig. S8). Second, the excitatory drive engages frequency-tuned intrinsic properties like HCN channels. This interplay shapes a broad signal into a specific output. Although multiple 5-HT receptors are expressed in the CN, as previously reported (36), we found that the gap detection deficits induced by activation of the 5-HT^{DRN→DCN} circuit were largely attenuated by pretreatment with MDL-11939, a selective antagonist for the 5-HT_{2A} receptor. These results suggest that blocking 5-HT_{2A} receptor partially reverses tinnitus-related behavior induced by activation of 5-HT^{DRN→DCN} circuit. These findings are consistent with our electrophysiological recordings (Fig. 2) and extend our previous finding demonstrating that ketanserin (a 5-HT_{2A} receptor antagonist) blocks 5-HT-evoked inward currents in the DCN (40), further supporting the involvement of 5-HT_{2A} receptors in this pathway. Although our data indicate that 5-HT_{2A} receptor is required for expression of tinnitus induced by the 5-HT^{DRN→DCN} circuit, the potential contribution of other 5-HT receptors cannot be ruled out. These results strongly suggest that hyperactivity within the 5-HT^{DRN→DCN} circuit represents a key mechanism underlying the generation of tinnitus.

Although SSRIs have been explored as a treatment for the negative quality-of-life associated with the tinnitus percept, some studies indicated that some patients report enhanced tinnitus after beginning SSRI regimen. However, there is a fundamental gap in our understanding how acute administration of SSRIs affects the activity of the auditory system at the cellular and system level. Based on our previous study (40) and the observation that acute citalopram administration induces tinnitus-related behavior in mice

(SI Appendix, Fig. S9 and Dataset S1M), we postulated that short-term SSRIs treatment results in hyperactivity in 5-HT^{DRN→DCN} circuit, subsequently triggering a tinnitus-related behavior. Such an explanation could help clarify the relationship between tinnitus and the clinical side effects of acute SSRIs treatment.

Several lines of evidence suggest that the aberrant activity within the 5-HT^{DRN→DCN} circuit is implicated in the perception of noise-induced tinnitus. The recent development of advanced serotonin biosensors, such as the G protein-coupled receptor activation-based (GRAB) 5-HT sensor, enables high-specificity, subsecond-resolution detection of extracellular 5-HT release in freely moving animals (50, 56). Utilizing the optimized GRAB 5-HT sensor (50), which exhibits enhanced sensitivity and spatiotemporal resolution compared to earlier GFP-based tools, we provide direct evidence that 5-HT release is enhanced in the DCN of noise-exposed mice with tinnitus-related behavior (Fig. 5). The observed increase in 5-HT levels within the DCN may at least partially result from the enhanced activity of 5-HT^{DRN→DCN} neurons. Consistent with this, GCaMP6f-based fiber photometry further demonstrated increased activity in 5-HT^{DRN→DCN} neurons of noise-exposed mice exhibiting tinnitus-related behavior (Fig. 5). These results indicate that hyperactivity in DRN serotonergic neurons projecting to the DCN drives the 5-HT dysregulation implicated in tinnitus-related behaviors. In addition, our histological analysis of c-fos expression demonstrated activation of 5-HT neurons in the DRN of noise-exposed mice exhibiting behavioral evidence of tinnitus (SI Appendix, Fig. S6). All these results suggest that hyperactivity of the 5-HT^{DRN→DCN} circuit is associated with noise-induced tinnitus in mice. More importantly, chemogenetic silencing 5-HT^{DRN→DCN} circuit significantly ameliorates the tinnitus-related behavior in mice (Fig. 6 and SI Appendix, Fig. S7), directly linking hyperactivity of 5-HT^{DRN→DCN} circuit to the behavioral manifestations of noise-induced tinnitus. Collectively, our results support the hypothesis that the 5-HT^{DRN→DCN} circuit plays a critical role in the modulation of noise-induced tinnitus perception.

In support of our hypothesis, Cransac et al. (1998) have shown that noise exposure in rats produces elevations in 5-HT and its metabolite 5-hydroxyindoleacetic acid (5-HIAA) within the CN (25). Furthermore, our previous study has demonstrated that loud noise exposure reduces SERT expression in the DCN of mice, which could produce prolonged 5-HT availability in the DCN, subsequently increase neural hyperactivity, and further highlights the potential role of 5-HT in modulating tinnitus-related behavior (26). Notably, upregulation of 5-HT levels within the central auditory system and activation of DRN 5-HT neurons also occur in another reversible type of tinnitus, specifically that induced by high doses of sodium salicylate (57). Taken together with previous studies (25, 26, 57), our data suggest that noise-induced tinnitus in mice is at least partially mediated by 5-HT release from 5-HT^{DRN→DCN} neurons. These findings extend our understanding of how noise-induced cochlear deafferentation drives homeostatic plasticity in the DCN. This plasticity appears to result from a combination of increased intrinsic excitability of fusiform cells and a decrease in inhibitory neurotransmission in the DCN (4).

Together, the induction of tinnitus through chemogenetic activation of the 5-HT^{DRN→DCN} circuit, coupled with the partial reversal of noise-induced tinnitus by inhibition of this circuit, underscores the critical role of hyperactivity within the 5-HT^{DRN→DCN} circuit as a key mechanism underlying tinnitus. This parallels established mechanisms in the central amygdala (CeA), where elevated 5-HT levels (hyperserotonergic states) and specific activation of 5-HT^{DRN→CeA} neurons, certain subpopulations of serotonergic neurons in the DRN, promote anxiety-like

behaviors (6, 12). We propose that noise exposure may contribute to hyperactivity in the DCN through a similar serotonergic mechanism: stress- or trauma-induced 5-HT release from some subpopulations of serotonergic neurons in the DRN could amplify neural excitability in the DCN, mirroring the CeA's role in anxiety. It would be interesting to investigate whether these subpopulations of serotonergic neurons also project to CeA. This convergence suggests a broader principle whereby hyperserotonergic states in distinct brain circuits mediate diverse neurobehavioral outcomes, e.g., anxiety in the CeA and tinnitus in the DCN.

In conclusion, our findings demonstrate that activating the 5-HT_{DRN→DCN} neural circuit reversibly drives DCN hyperexcitability and tinnitus-related behavior via a mechanism of dynamic circuit plasticity, rather than permanent pathology. This effect is largely mediated by 5-HT_{2A} receptor activation, which disrupts local excitation-inhibition balance through short-term adaptation. This indicates that certain forms of tinnitus reflect a neuromodulator-triggered, dynamic state, in which 5-HT acts as a plastic trigger, reconfiguring auditory circuits into a hyperexcitable mode that generates or sustains the tinnitus percept. This work identifies a discrete serotonergic circuit underlying tinnitus in hyperserotonergic states and establishes a circuit-based foundation for targeted therapies addressing sensory processing disorders.

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Materials and Methods

Detailed materials and methods are provided in *SI Appendix*. Briefly, tinnitus was assessed in mice using the Gap Pre-Pulse Inhibition of the Acoustic Startle (GPIAS) paradigm, which measures the detection of a silent gap within background sounds of varying frequencies. To dissect the underlying neural circuitry, we employed retrograde and anterograde viral tracing, fiber photometry, and optogenetic/chemogenetic manipulations to selectively target and modulate the 5-HT_{DRN→DCN} neural circuit. The functional impact of this circuit on neuronal activity was validated through in vivo extracellular single-unit recordings and brain slice electrophysiology in DCN neurons. Control experiments, including auditory brainstem response (ABR) tests and prepulse inhibition (PPI), were conducted to rule out hearing loss and confirm the specificity of the observed behavioral deficits.

Data, Materials, and Software Availability. All study data are included in the article and/or [supporting information](#).

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